吉林大学

DB2 实验报告

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姓名	柳湛博	学号	55241019
实验项目	5. Backup and recovery		
实验性质	☐演示性实验 ☐验证性实验 ☑操作性实验 ☐综合性实验		
实验地点	A108	机器编号	10
指导教师	刘雪洁	实验时间	2026 年 9 月 28 日 20 时 4 分

一、实验综述

1. 实验目的及要求

This exercise is an online lab where the database MUSICDB will be used for backup and recovery. During this lab the students will prepare the database for archival logging, create a backup and restore and recover the database to the current as well as to a prior point in time.

At the end of the lab, students should be able to:

- Create backup images of a database and of table spaces
- Restore a database from a backup image
- Roll forward a database or table spaces to perform full recovery
- Perform actions to reactivate databases in exception states

2. 实验设备、软件

PC, windows XP Professional, DB2 9 Express-c

二、实验过程（实验步骤、记录、数据、分析）

Section 1 - Determining Log Configurations

1. Before beginning this lab, you need to make sure some of your database configuration parameters are set to the values required by this lab. Do this by running the script recovrst.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "update db cfg for lzb_1019 usin
g logretain no dbheap 960 logfilsiz 6 softmax 250 logprimary 3 logsecond 2"
DB20000I  UPDATE DATABASE CONFIGURATION 命令成功完成。
SQL1363W  未动态更改为立即修改而提交的一个或多个参数。对于这些配置参数，必须在所

有应用程序都与此数据库断开连接之后，更改才会生效。

D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 force application all
DB20000I  FORCE APPLICATION 命令成功完成。
DB21024I  此命令为异步的，可能未能立即生效。


D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 terminate
DB20000I  TERMINATE 命令成功完成。

D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>
```

2. Connect to MUSICDB to activate the database.
3. Examine your db cfg file for MUSICDB.
How large are the log files? Is this amount in pages or bytes?
Show your work below by printing screen and answer the question.



6

Pages

4. How many primary log files will be allocated? When will these files be allocated?

3
在数据库被激活时（例如第一次连接到该数据库时）进行分配。

5. How many secondary log files will be allocated? When will these files be allocated?
2 (由 `LOGSECOND` 参数决定)。
当主日志文件（Primary log files）写满，需要更多日志空间时动态分配。

6. For which type of logging is your database currently configured (log retention or circular)? What parameters provide this information?

循环日志（Circular logging）。

Log retain for recovery status (LOGRETAIN) = OFF` 且 `User exit for logging status (USEREXIT) = OFF`。

7. Which types of recovery are supported by this type of logging — Crash, Version, or Roll forward?

崩溃恢复（Crash recovery）和版本恢复（Version recovery，即从脱机备份中恢复整个数据库）。不支持前滚恢复（Roll forward）。

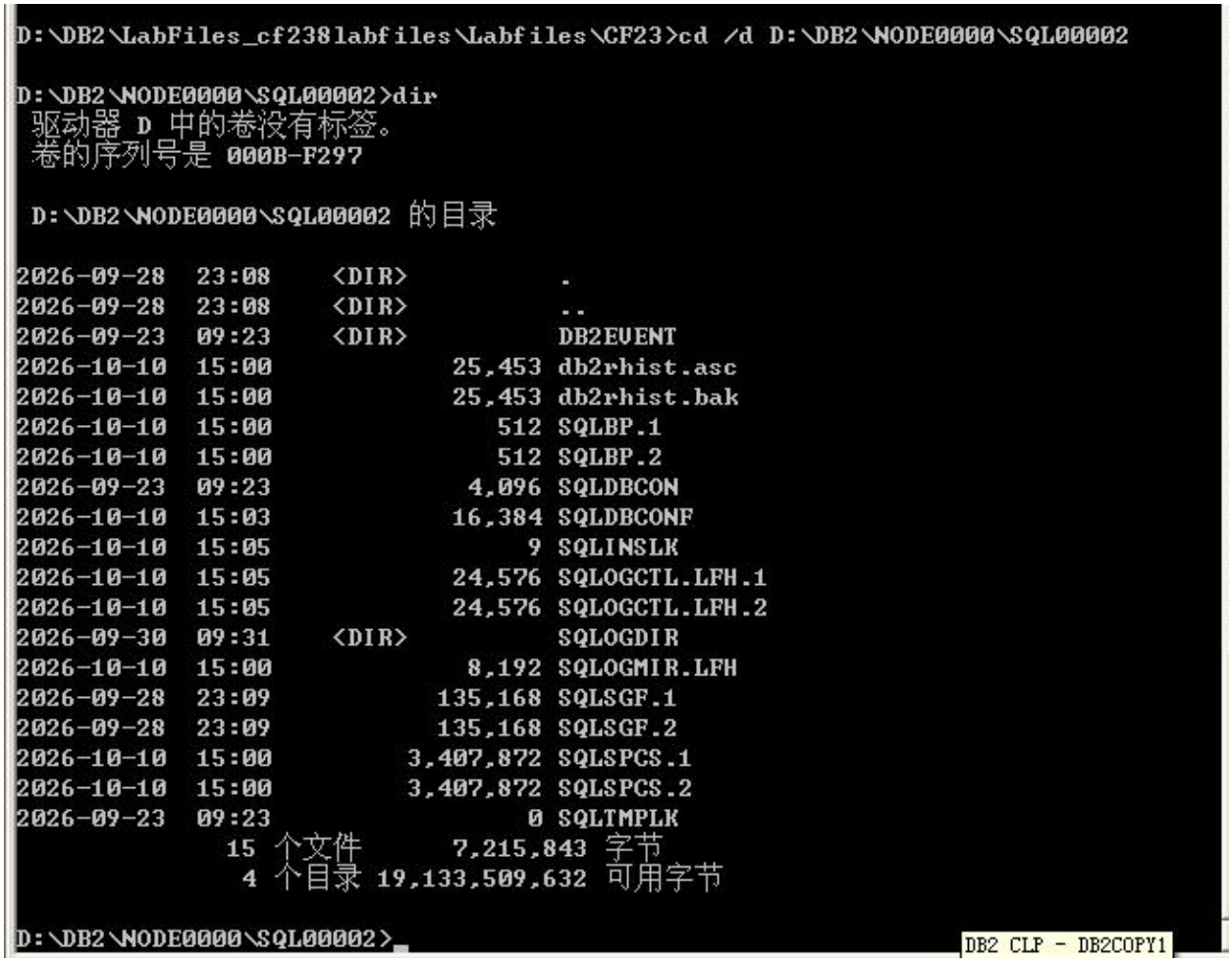
8. Where do the log files reside?

根据 `db cfg` 中的 `Path to log files` 参数

在 `D:\DB2\NODE0000\SQL00001\SQLOGDIR\`

9. Assume the database manager was active. A short interruption of power made it necessary to perform a crash recovery. Will manual intervention be required to cause this crash recovery? What configuration parameter relates to this issue?
NO
Auto restart enabled (AUTORESTART) = ON`。
是否需要人工干预：不需要（No manual intervention is required）。
相关配置参数：`AUTORESTART`（自动重新启动标志）。当该参数设定为 `ON` 时，一旦发生意外断电重启，当应用程序下次连接数据库时，DB2 引擎将全自动检测并静默完成崩溃恢复，使数据库重回一致性状态。

10. From your Command Window session, change your current directory to the log path, but exclude the SQLOGDIR directory from your command.
Note: Do NOT EDIT these files!
Show your work below by printing screen.



11. Determine the contents of this directory.
What are T0000000, T0000001, T0000002?
它们是当前数据库内部表空间（Table Spaces）的数据容器或元数据控制目录/文件。在 DB2 底层架构中，每个以 `T` 开头的对象编号对应数据库中特定表空间 ID（例如系统编目表空间、临时表空间或系统管理表空间的内部存储句柄）。
12. Is it appropriate to keep the log files in the same path? (Consider the type of recovery currently supported in your answer.)
不合理。为了更好的性能（避免 I/O 争用）和在磁盘发生故障时进行恢复，日志文件通常应当与数据库数据文件放置在不同的物理磁盘上。

13. Now, list the contents of SQLOGDIR.
Show your work below by printing screen.



14. How many log files are in this directory? Does this match what you expected considering your configuration file? Why or why not?

3 个
是的，匹配。因为参数 `LOGPRIMARY` 被设置为 3，且数据库已经激活，

所以预先分配了 3 个主日志文件。

15. Increase the quantity in the STOCK table by 1 but do not commit the update. If you are not certain how to turn off auto-commit, check the solutions.
Show your work below by printing screen.



16. If you encounter an error, go to the next question. If you did NOT encounter any error, decrease the quantity in the STOCK table by 1.
Show your work below by printing screen.

17. What did you observe? What was the SQL code you received?
更新操作被系统拒绝，提示事务日志已满；
收到的 SQL 代码：SQL0964C（SQLSTATE=57011）。

18. Issue a command to retrieve help on the SQL code. Why did the statement fail?
Show your work below by printing screen.

SQL0964C 数据库的事务日志已满。

说明:

已使用事务日志中的所有空间。

如果使用具有辅助日志文件的循环日志，那么尝试分配和使用这些日志。当文件系统没有更多空间时，不能使用辅助日志。

如果使用归档日志，那么文件系统不提供空间来包含新日志文件。

当数据库几乎已用完所有可能的日志序号时，也会返回此消息。数据库管理器使用称为日志序号（LSN）的唯一标识来标识数据库日志记录。日志序号是范围介于 0x0000 0000 0000 到 0xFFFF FFFF FFFF 之间的 6 字节十六进制数。当事务导致写入数据库日志时，数据库将使用新的 LSN 值。

应用程序无法执行任何导致数据库写入日志记录的事务。

无法处理该语句。

用户响应:

在接收到此消息（SQLCODE）时，执行 COMMIT 或 ROLLBACK，或再次尝试执行该操作。

如果并发应用程序正在更新数据库，那么再次尝试执行该操作。当另一个应用程序完成事务时，可能释放日志空间。

发出更频繁的落实操作。如果事务还未落实，那么当落实事务时，可能会释放日志空间。设计应用程序时，应考虑何时落实更新事务，以防止发生日志已满的情况。

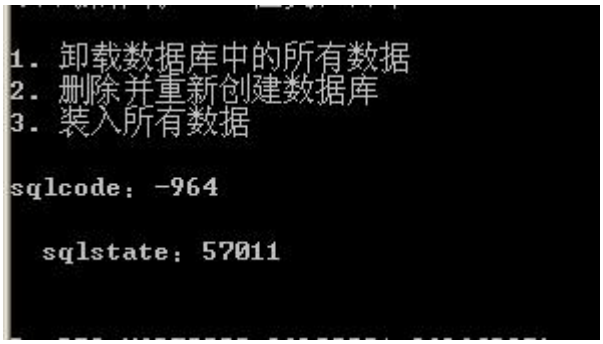
如果发生死锁，那么更频繁地对它们进行检查。这可以通过减小数据库配置参数 DLCHKTIME 来实现。这将检测到死锁，并且很快解决（通过 ROLLBACK），这将释放日志空间。

如果经常发生这种情况，那么增大数据库配置参数以允许更大的日志文件。更大的日志文件需要更多空间，但是减少了应用程序再次尝试执行该操作的需要。可能需要调整的事务配置参数是 LOGFILSIZ、LOGPRIMARY 和 LOGSECOND。有关事务日志的更多信息，请使用诸如"事务日志"之类的短语在 DB2 信息中心中进行搜索。

如果正在安装样本数据库，那么删除它并再次安装样本数据库。

如果由于数据库几乎用完所有可能的 LSN 值而返回此消息，那么您可以通过执行以下操作将 LSN 值复位为零:

1. 卸载数据库中的所有数据
2. 删除并重新创建数据库
3. 装入所有数据



19. Since the unit of work could not complete successfully, issue a rollback.
Show your work below by printing screen.



20. From your telnet session, determine the number of log files allocated at this point. How many are there?

5 个。

21. What accounts for the additional files?

当 3 个主日志文件写满后，由于事务没有提交，数据库管理器根据 `LOGSECOND` 参数分配了 2 个额外的辅助日志文件。

22. Change to your home directory in your Command window.



23. From your telnet session, create a directory that will contain backups of your database: mkdir restore



Section 2 - Backup/Restore Support with Circular Logging

1. Connect to your musicdb database from the new command window.

Show your work below by printing screen.

```
D:\>db2 connect to lzb_1019

   数据库连接信息

数据库服务器          = DB2/NT 9.7.0
SQL 授权标识          = LZB_1019
本地数据库别名        = LZB_1019


D:\>_
```

2. Attempt to perform an online backup of your MUSICDB database to the directory of C:\restore. Were you successful in trying to specify an online backup? Explain why or why not.

Show your work below by printing screen and answer the question.

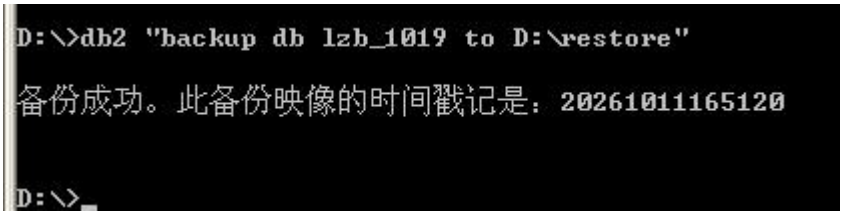
```
D:\>db2 "backup db lzb_1019 online to D:\restore"
SQL2413N  不允许联机备份，因为没有激活用于前滚的日志保留或用户出口，或者是因为备

份暂挂条件正在对数据库起作用。

D:\>_
```

无法成功执行联机备份。原因分析：联机备份（Online Backup）允许用户在备份期间并发读写数据库，这意味着备份映像必须结合备份期间生成的事务日志才能恢复到一致性状态。因此 DB2 强制规定联机备份必须在归档日志（支持前滚恢复）模式下运行。当前数据库配置为循环日志（`LOGRETAIN=OFF`），不支持前滚恢复，因而直接拒绝联机备份。

3. Attempt to perform an offline backup instead.
Show your work below by printing screen.



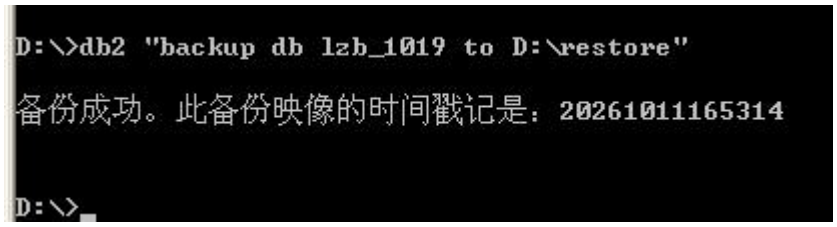
4. Did the backup complete successfully? What did the message say?
是的，备份直接成功了。提示信息为：“备份成功。此备份映像的时间戳记是: 20261011165120”。

5. In order to take an offline backup, do you think that exclusive use of the database is required? Who is currently connected to the database?
是的，脱机备份需要独占访问。因为我当前没有任何应用连接到该数据库，所以获得了独占访问权并备份成功。

6. Resolve the apparent problem by forcing all applications to end.
Show your work below by printing screen.



7. Now try your backup again.
Show your work below by printing screen.



8. Was the backup successful this time?
是的

9. The timestamp for a backup image is necessary for restore if you are using the RESTORE command rather than the GUI DB2 Control Center, and multiple images reside in a given source.
When the backup is targeted to disk, the name of the file itself will reflect the timestamp. When tape or TSM managed devices are used, the header in the backup image will contain this information.

10. Look at the backup information through the LIST BACKUP command. Remember that you have a previous backup from an early lab (LOAD lab). Issue the list backup command with the ALL keyword.

Show your work below by printing screen.

```
D:\>db2 list history backup all for lzb_1019

                    列示 lzb_1019 的历史记录文件

匹配的文件条目数 = 4

 Op 对象 时间戳记+序列      类型 设备 最早日志     当前日志     备份标识
 -- --- ------------------ ---- --- ------------ ------------ --------------
  B  D  20261011092935001   F    D  S0000000.LOG S0000000.LOG
 ------------------------------------------------------------------------
  包含 10 表空间:

  00001 SYSCATSPACE
  00002 USERSPACE1
  00003 DMS01
  00004 DMS02
  00005 DMS03
  00006 DMS04
  00007 DMS05
  00008 DMS06
  00009 SMS01
  00010 SMS02
 ------------------------------------------------------------------------
    注释: DB2 BACKUP LZB_1019 OFFLINE
 开始时间: 20261011092935
   结束时间: 20261011092938
     状态: A
 ------------------------------------------------------------------------
  EID: 12 位置: D:\backup

 Op 对象 时间戳记+序列      类型 设备 最早日志     当前日志     备份标识
 -- --- ------------------ ---- --- ------------ ------------ --------------
  B  D  20261011165026000   N       S0000000.LOG
 ------------------------------------------------------------------------


 ------------------------------------------------------------------------
    注释: DB2 BACKUP LZB_1019 ONLINE
 开始时间: 20261011165026
   结束时间: 20261011165027
     状态: A
 ------------------------------------------------------------------------
  EID: 17 位置:

SQLCA 信息
```

```
sqlcaid : SQLCA     sqlcabc: 136   sqlcode: -2413   sqlerrml: 0

 sqlerrmc:
sqlerrp : sqlubIni
sqlerrd : (1) 0                (2) 0                (3) 0
          (4) 0                (5) 0                (6) 0
sqlwarn : (1)      (2)      (3)      (4)       (5)       (6)
          (7)      (8)      (9)      (10)      (11)
 sqlstate:

Op 对象 时间戳记+序列     类型 设备 最早日志     当前日志     备份标识
-- --- ------------------ ---- --- ------------ ------------ --------------
 B  D  20261011165120001   F    D  S0000000.LOG S0000000.LOG
--------------------------------------------------------------------------------
 包含 10 表空间:

 00001 SYSCATSPACE
 00002 USERSPACE1
 00003 DMS01
 00004 DMS02
 00005 DMS03
 00006 DMS04
 00007 DMS05
 00008 DMS06
 00009 SMS01
 00010 SMS02
--------------------------------------------------------------------------------
   注释: DB2 BACKUP LZB_1019 OFFLINE
开始时间: 20261011165120
  结束时间: 20261011165121
    状态: A
--------------------------------------------------------------------------------
 EID: 18 位置: D:\restore


Op 对象 时间戳记+序列     类型 设备 最早日志     当前日志     备份标识
-- --- ------------------ ---- --- ------------ ------------ --------------
 B  D  20261011165314001   F    D  S0000000.LOG S0000000.LOG
--------------------------------------------------------------------------------
 包含 10 表空间:

 00001 SYSCATSPACE
 00002 USERSPACE1
 00003 DMS01
 00004 DMS02
 00005 DMS03
 00006 DMS04
 00007 DMS05
 00008 DMS06
 00009 SMS01
 00010 SMS02
```



11. Is the timestamp available as part of this information?
是的。

12. Record the start timestamp of the backup.
20261011165315

13. How many table spaces are included in this backup?
10

14. Instead of using the keyword ALL, it is possible to restrict the output of the LIST BACKUP command via a timestamp specification or by object name specification.
Issue a request to get just the backups for today.
You may wish to request online help for this command.
Show your work below by printing screen.

```
                    列示 lzb_1019 的历史记录文件

匹配的文件条目数 = 4

 Op 对象 时间戳记+序列      类型 设备 最早日志     当前日志     备份标识
 -- --- ------------------ ---- --- ------------ ------------ --------------
  B  D  20261011092935001   F    D  S0000000.LOG S0000000.LOG
 ------------------------------------------------------------------------------
  包含 10 表空间:

  00001 SYSCATSPACE
  00002 USERSPACE1
  00003 DMS01
  00004 DMS02
  00005 DMS03
  00006 DMS04
  00007 DMS05
  00008 DMS06
  00009 SMS01
  00010 SMS02
 ------------------------------------------------------------------------------
    注释: DB2 BACKUP LZB_1019 OFFLINE
 开始时间: 20261011092935
   结束时间: 20261011092938
     状态: A
 ------------------------------------------------------------------------------
  EID: 12 位置: D:\backup

 Op 对象 时间戳记+序列      类型 设备 最早日志     当前日志     备份标识
 -- --- ------------------ ---- --- ------------ ------------ --------------
  B  D  20261011165026000   N       S0000000.LOG
 ------------------------------------------------------------------------------


 ------------------------------------------------------------------------------
    注释: DB2 BACKUP LZB_1019 ONLINE
 开始时间: 20261011165026
   结束时间: 20261011165027
     状态: A
 ------------------------------------------------------------------------------
  EID: 17 位置:

SQLCA 信息

 sqlcaid : SQLCA     sqlcabc: 136   sqlcode: -2413   sqlerrml: 0

  sqlerrmc:
 sqlerrp : sqlubIni
 sqlerrd : (1) 0               (2) 0               (3) 0
           (4) 0               (5) 0               (6) 0
 sqlwarn : (1)      (2)      (3)      (4)        (5)       (6)
```

```
          (4) 0                (5) 0                (6) 0
sqlwarn : (1)      (2)      (3)      (4)       (5)       (6)
          (7)      (8)      (9)      (10)      (11)
 sqlstate:

Op 对象 时间戳记+序列      类型 设备 最早日志     当前日志     备份标识
-- --- ------------------ ---- --- ------------ ------------ --------------
 B  D  20261011165120001   F    D  S0000000.LOG S0000000.LOG
--------------------------------------------------------------------------------
 包含 10 表空间:

 00001 SYSCATSPACE
 00002 USERSPACE1
 00003 DMS01
 00004 DMS02
 00005 DMS03
 00006 DMS04
 00007 DMS05
 00008 DMS06
 00009 SMS01
 00010 SMS02
--------------------------------------------------------------------------------
   注释: DB2 BACKUP LZB_1019 OFFLINE
开始时间: 20261011165120
  结束时间: 20261011165121
    状态: A
--------------------------------------------------------------------------------
 EID: 18 位置: D:\restore


Op 对象 时间戳记+序列      类型 设备 最早日志     当前日志     备份标识
-- --- ------------------ ---- --- ------------ ------------ --------------
 B  D  20261011165314001   F    D  S0000000.LOG S0000000.LOG
--------------------------------------------------------------------------------
 包含 10 表空间:

 00001 SYSCATSPACE
 00002 USERSPACE1
 00003 DMS01
 00004 DMS02
 00005 DMS03
 00006 DMS04
 00007 DMS05
 00008 DMS06
 00009 SMS01
 00010 SMS02
--------------------------------------------------------------------------------
   注释: DB2 BACKUP LZB_1019 OFFLINE
开始时间: 20261011165314
  结束时间: 20261011165315
    状态: A
--------------------------------------------------------------------------------
 EID: 19 位置: D:\restore                    DB2 CLP - DB2COPY1
```

15. The history information regarding backups is maintained by the database manager and can become extensive if it is never removed. Use the ? command to invoke the online help facility to identify a command that can be used to manage the amount of history information retained. Write the name of the command you believe would give you the capability to eliminate old history information.

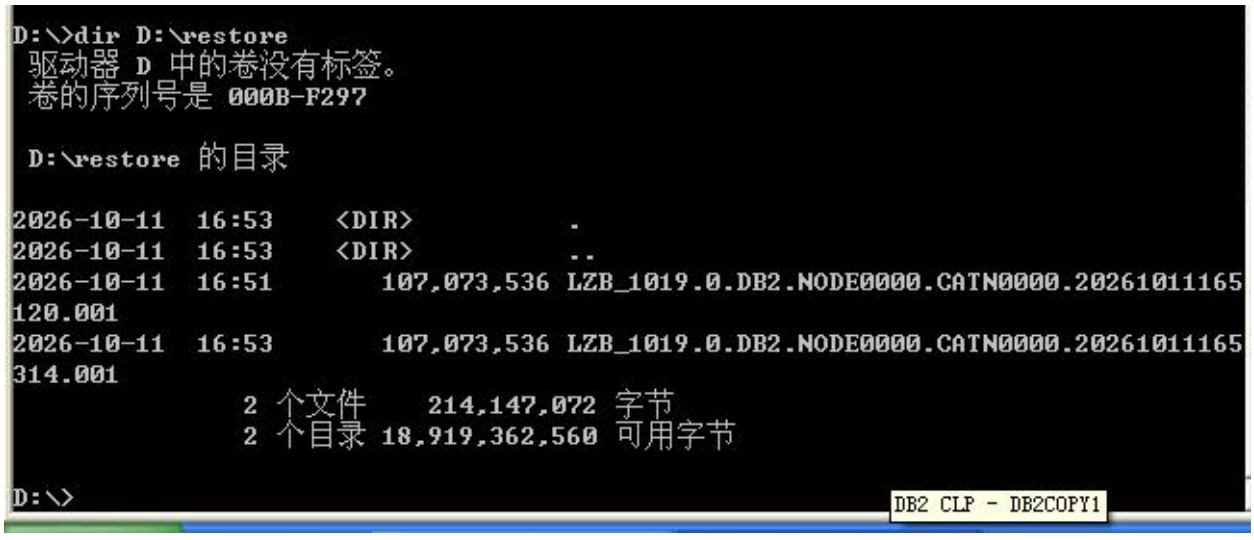
Show your work below by printing screen.

```
C:\>db2 "? prune history"
PRUNE HISTORY timestamp [WITH FORCE OPTION] [AND DELETE]
```

PRUNE HISTORY

16. From your Command Window, examine the file naming convention used for your backup.

Show your work below by printing screen.



17. The backup image reflects the same data that is currently in your MUSICDB database since you have not issued any SQL to change data since the backup was taken. Select the artist's name and album titles for artist number (ARTNO) 77.
Show your work below by printing screen.

```
D:\>db2 connect to lzb_1019

   数据库连接信息

 数据库服务器          = DB2/NT 9.7.0
 SQL 授权标识          = LZB_1019
 本地数据库别名        = LZB_1019


D:\>db2 "select a.name, b.title from lzb_1019.artists a, lzb_1019.albums b where
 a.artno=b.artno and a.artno=77"

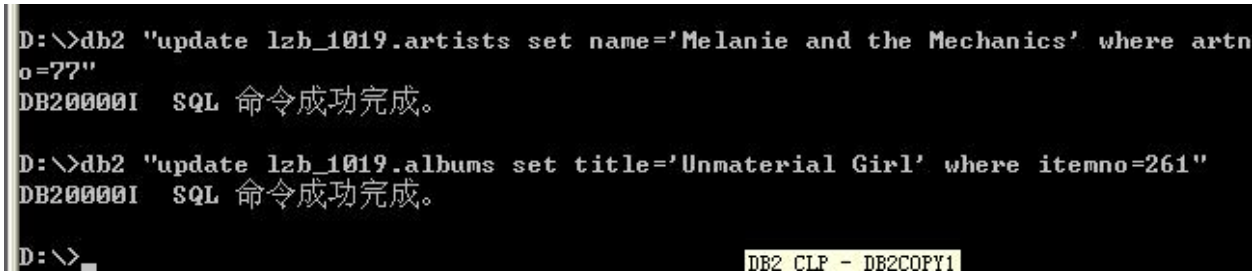
NAME                                               TITLE
-------------------------------------------------- ------------------------------
----------------------
Livid Ids                                          The First 13

Livid Ids                                          Live at the Psychidelly


  2 条记录已选择。
```

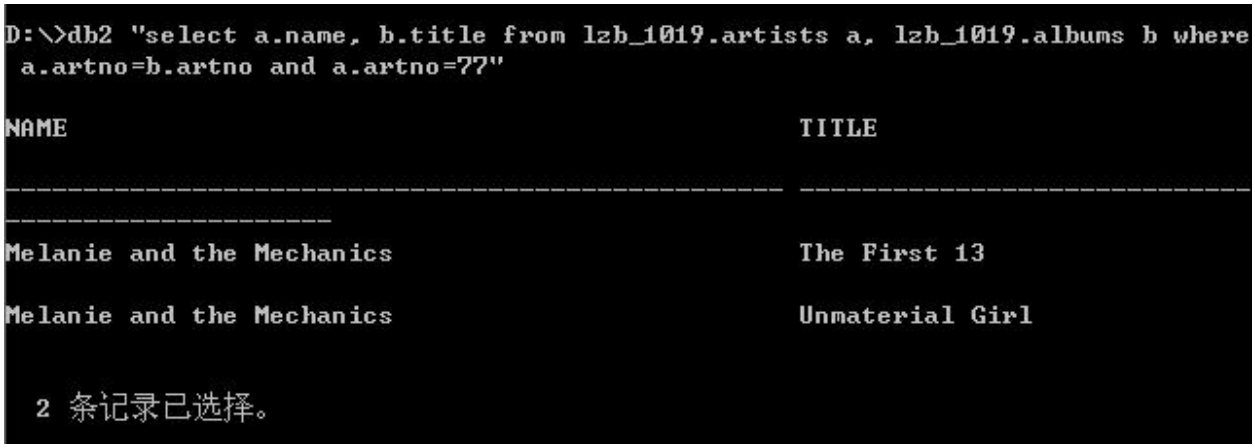
18. Change the name of the artist 77 to “Melanie and the Mechanics” and the album title of ITEMNO 261 to “Unmaterial Girl”.

Show your work below by printing screen.



19. Execute a join between ARTISTS and ALBUMS for ARTNO 77 and document the data as it now appears. This reflects a change that occurred after your backup.

Show your work below by printing screen.



20. Assume the disk containing the MUSICDB database failed. You now have the task to restore the backup image that was made.

Show your work below by printing screen.

```
D:\>db2 force application all
DB20000I  FORCE APPLICATION 命令成功完成。
DB21024I  此命令为异步的，可能未能立即生效。


D:\>db2 terminate
DB20000I  TERMINATE 命令成功完成。

D:\>db2 "restore db lzb_1019 from D:\restore"
SQL2522N  多个备份文件与为备份的数据库映像提供的时间戳记值相匹配。
```

21. First make sure there are no applications are connected to the database.

Show your work below by printing screen.

```
D:\>db2 force application all
DB20000I  FORCE APPLICATION 命令成功完成。
DB21024I  此命令为异步的，可能未能立即生效。


D:\>db2 terminate
DB20000I  TERMINATE 命令成功完成。

D:\>db2 "restore db lzb_1019 from D:\restore"
SQL2522N  多个备份文件与为备份的数据库映像提供的时间戳记值相匹配。
```

22. Restore the database from the C:\restore directory.

Show your work below by printing screen.

```
D:\>db2 force application all
DB20000I  FORCE APPLICATION 命令成功完成。
DB21024I  此命令为异步的，可能未能立即生效。


D:\>db2 terminate
DB20000I  TERMINATE 命令成功完成。

D:\>db2 "restore db lzb_1019 from D:\restore"
SQL2522N  多个备份文件与为备份的数据库映像提供的时间戳记值相匹配。
```

23. Connect to MUSICDB and execute the join between ARTISTS and ALBUMS for ARTNO 77.

Show your work below by printing screen.

```
D:\>db2 connect to lzb_1019

   数据库连接信息

 数据库服务器          = DB2/NT 9.7.0
 SQL 授权标识          = LZB_1019
 本地数据库别名        = LZB_1019


D:\>db2 "select a.name, b.title from lzb_1019.artists a, lzb_1019.albums b where
 a.artno=b.artno and a.artno=77"

NAME                                               TITLE
-------------------------------------------------- ------------------------------
--------------------
Melanie and the Mechanics                          The First 13

Melanie and the Mechanics                          Unmaterial Girl


  2 条记录已选择。
```

24. Does the data reflect the updates made, or is the original data restored?
显示的是备份时的原始数据，我们在备份之后做的修改已经丢失。

25. Assume that you are not in a lab environment. You made your backup three days ago and thousands of transactions have since occurred. Would your current recovery strategy be acceptable?
不可接受。因为自备份之后发生的所有成千上万笔事务修改都会在恢复时完全丢失，且无法通过前滚日志找回这些修改记录。

Section 3 - Backup/Restore Support with Archive Logging

1. Update the database configuration file to indicate that log retention logging is to be used (recovery).

Show your work below by printing screen.

```
D:\>db2 "update db cfg for lzb_1019 using logretain recovery"
DB20000I  UPDATE DATABASE CONFIGURATION 命令成功完成。
SQL1363W  未动态更改为立即修改而提交的一个或多个参数。对于这些配置参数，必须在所

有应用程序都与此数据库断开连接之后，更改才会生效。

D:\>
```

2. Some changes to the db cfg file (including changes to LOGRETAIN) do not take affect until the database is deactivated and becomes active again. Since the database was activated by the first user connect, all applications must end to deactivate the database. Try issuing a DB2 terminate command.

Show your work below by printing screen.

```
D:\>db2 terminate
DB20000I  TERMINATE 命令成功完成。

D:\>db2 force application all
DB20000I  FORCE APPLICATION 命令成功完成。
DB21024I  此命令为异步的，可能未能立即生效。

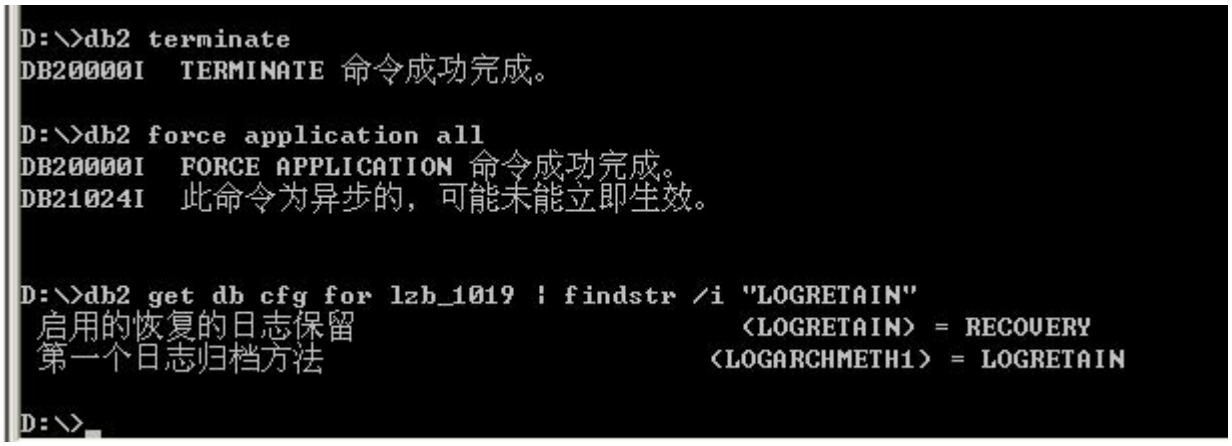

D:\>db2 get db cfg for lzb_1019 | findstr /i "LOGRETAIN"
 启用的恢复的日志保留                         (LOGRETAIN) = RECOVERY
 第一个日志归档方法                          (LOGARCHMETH1) = LOGRETAIN

D:\>_
```

3. To find out for sure if log retention logging is in effect now, look at the Log

retain status indicator in the db cfg.

Show your work below by printing screen.



4. Did the terminate command change anything concerning the parameters?

LOGRETAIN` 的状态可能仍未完全生效，因为可能还有其他连接未断开。

5. Try forcing all applications off the database.

Show your work below by printing screen.



6. Did the Log retain status indicator value change to RECOVERY?

Show your work below by printing screen and answer the question

是的，完全断开连接后配置生效，变成了 RECOVERY。

7. Connect to musicdb database. Was the connection successful? What does the SQL message indicate? Why does the database manager require a new backup?

Show your work below by printing screen and answer the question



消息说明：数据库当前处于 备份暂挂（Backup Pending） 保护状态。
当数据库从循环日志切换为归档日志后，日志机制发生了本质蜕变。DB2 必须强制要求管理员做一次全量脱机基准备份（Baseline Backup），以此作为未来所有增量日志前滚重放的“原点”。在获得该基线前，数据库严禁读写。

8. Did the attempt to connect to the database cause log retention to be activated?
Show your work below by printing screen and answer the question

是的，更改配置并完全断开连接后，数据库激活了归档日志模式，并立即将其置于 backup pending 状态。

9. Create an offline backup of your database and record the timestamp.
Show your work below by printing screen and answer the question.



10. Complete a series of SQL statements to change data in the ARTISTS table and record points in time associated with these changes.

a.Change the name of ARTNO 77 to "Melanie and the Mechanics".
b.Record the time.
c.Change the name of ARTNO 77 to "Bill Pellett and the Comets".
d.Record the time.

Show your work below by printing screen.

```
D:\>db2 connect to lzb_1019

   数据库连接信息

 数据库服务器          = DB2/NT 9.7.0
 SQL 授权标识          = LZB_1019
 本地数据库别名        = LZB_1019


D:\>db2 "update lzb_1019.artists set name='Melanie and the Mechanics' where artn
o=77"
DB20000I  SQL 命令成功完成。

D:\>db2 "values current timestamp"

1
--------------------------
2026-10-11-18.09.23.425000

  1 条记录已选择。


D:\>db2 "update lzb_1019.artists set name='Bill Pellett and the Comets' where ar
tno=77"
DB20000I  SQL 命令成功完成。

D:\>db2 "values current timestamp"

1
--------------------------
2026-10-11-18.09.29.659000

  1 条记录已选择。
```

11. View the db cfg again. What is the value of the first active log file?
Show your work below by printing screen and answer the question.

```
D:\>db2 get db cfg for lzb_1019 | findstr /i "First active log file"
 目录高速缓存大小 (4KB)                  (CATALOGCACHE_SZ) = 300
 日志缓冲区大小 (4KB)                           (LOGBUFSZ) = 256
 日志文件大小 (4KB)                            (LOGFILSIZ) = 6
 主日志文件的数目                             (LOGPRIMARY) = 3
 辅助日志文件的数目                            (LOGSECOND) = 2
 已更改的至日志文件的路径                      (NEWLOGPATH) =
 日志文件路径                                               = D:\DB2\NODE0000\SQL00
003\SQLOGDIR\
 溢出日志路径                            (OVERFLOWLOGPATH) =
 镜像日志路径                              (MIRRORLOGPATH) =
 首个活动日志文件                                           = S0000000.LOG
 磁盘上已满的块日志                      (BLK_LOG_DSK_FUL) = NO
 非记录块操作                            (BLOCKNONLOGGED) = NO
 事务使用的最大主日志空间的百分比          (MAX_LOG) = 0
 1 个活动 UOW 的活动日志文件的数目         (NUM_LOG_SPAN) = 0
 启用的恢复的日志保留                        (LOGRETAIN) = RECOVERY
 第一个日志归档方法                        (LOGARCHMETH1) = LOGRETAIN
 logarchmeth1 的选项                        (LOGARCHOPT1) =
 第二个日志归档方法                        (LOGARCHMETH2) = OFF
 logarchmeth2 的选项                        (LOGARCHOPT2) =
 在索引构建期间记录页                     (LOGINDEXBUILD) = OFF
```

12. Restore your backup.

Show your work below by printing screen.

```
D:\>db2 terminate
DB20000I  TERMINATE 命令成功完成。

D:\>db2 "restore db lzb_1019 from D:\restore taken at 20261011165315"
SQL2542N  根据所提供的源数据库别名 "LZB_1019" 和时间戳记
"20261011165315", 找不到与数据库映像文件相匹配的文件。

D:\>db2 "restore db lzb_1019 from D:\restore taken at 20261011180830"
SQL2539W  警告! 正在复原至与备份映像数据库相同的现有数据库。数据库文件将被删除。


想要继续吗? (y/n) y
DB20000I  RESTORE DATABASE 命令成功完成。

D:\>
```

13. Did the restore fail?

没有

14. If the directory containing the backup image you wish to restore contains other images for the same database alias, you must specify which image to use with the TAKEN AT option on the restore command. If the previous restore failed, issue the restore command indicating which backup you wish to be restored via the TAKEN AT option.
Show your work below by printing screen.



15. Earlier in this lab, you attempted to connect to the database when it was in a backup pending status. You received a message during the connect to indicate this status. There is another method to determine the state of the database. Examine the db cfg again. Check the Recovery Status values.
Show your work below by printing screen.

```
D:\>db2 get db cfg for lzb_1019 | findstr "暂挂"
 备份暂挂                                                    = NO
 前滚暂挂                                                    = DATABASE
 复原暂挂                                                    = NO

D:\>
```

16. Is the database in any pending state?

是的，数据库当前正处于“前滚暂挂（Rollforward pending）”状态，其状态值为 DATABASE（全数据库级前滚暂挂）。

17. This state will occur if a restore is done to an online backup or a restore is done to an offline backup without specifying the WITHOUT ROLLING FORWARD phrase.
Show your work below by printing screen.

```
D:\>db2 "rollforward db lzb_1019 to end of logs and stop"

                              前滚状态

 输入数据库别名                        = lzb_1019
 节点数已返回状态                    = 1

 节点号                                = 0
 前滚状态                                    = 未暂挂
 下一个要读取的日志文件              =
 已处理的日志文件                    = S0000000.LOG - S0000000.LOG
 上次落实的事务                      = 2026-10-11-10.09.28.000000 UTC

DB20000I  ROLLFORWARD 命令成功完成。
```

18. Check the db cfg again to see if the Rollforward pending status was removed.

Was it?

Show your work below by printing screen and answer the question.

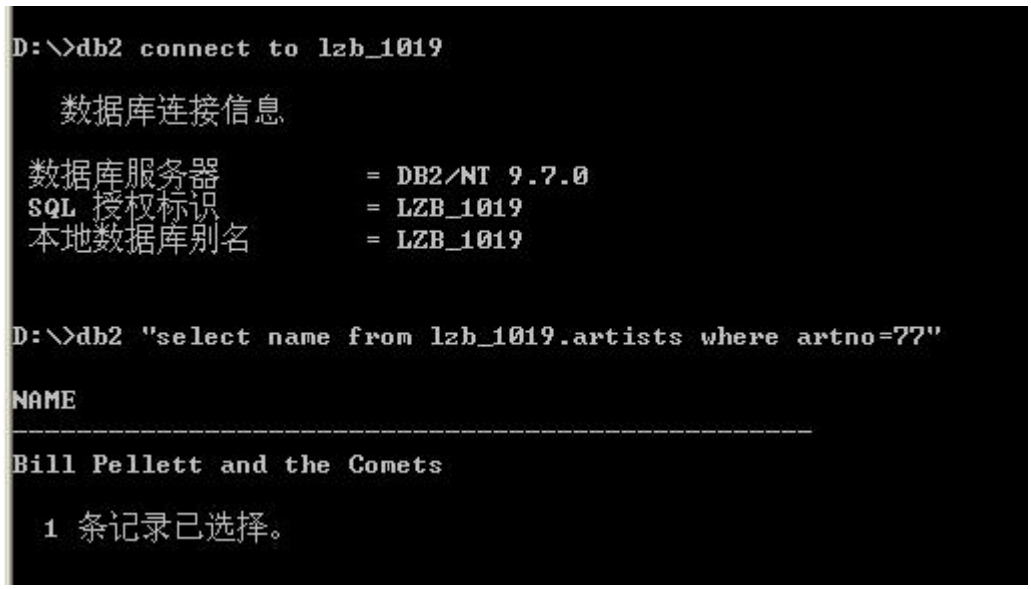


是的

19. Connect to MUSICDB database.

Check the contents of the row you updated earlier (ARTNO 77).

Show your work below by printing screen.



20. What is the value of the name returned? Why?

Bill Pellett and the Comets`。

因为我们启用了归档日志，并且进行了前滚恢复（Rollforward）操作。前滚操作读取了事务日志并重播了自备份以来的所有更改操作，从而恢复到了系统故障前的最新状态。

21. Even though the database manager is configured for roll forward recovery, you are not required to apply log records.
Restore the database backup and indicate that you do not want to roll forward.
Show your work below by printing screen.

```
D:\>db2 force application all
DB20000I  FORCE APPLICATION 命令成功完成。
DB21024I  此命令为异步的，可能未能立即生效。


D:\>db2 terminate
DB20000I  TERMINATE 命令成功完成。

D:\>db2 "restore db lzb_1019 from D:\restore taken at 20261011yyyyyy without rol
ling forward"
SQL2508N  “数据库复原 ”时间戳记参数 "20261011yyyyyy" 无效。

D:\>db2 "restore db lzb_1019 from D:\restore taken at 20261011180830 without rol
ling forward"
SQL2539W  警告！正在复原至与备份映像数据库相同的现有数据库。数据库文件将被删除。


想要继续吗？ (y/n) y
DB20000I  RESTORE DATABASE 命令成功完成。
```

22. Check the pending status of your database by looking at the db cfg.
Is the database in any pending state? Did the value returned to you match what you expected?
Show your work below by printing screen.

```
D:\>db2 get db cfg for lzb_1019 | findstr "暂挂"
 备份暂挂                                                    = NO
 前滚暂挂                                                    = NO
 复原暂挂                                                    = NO

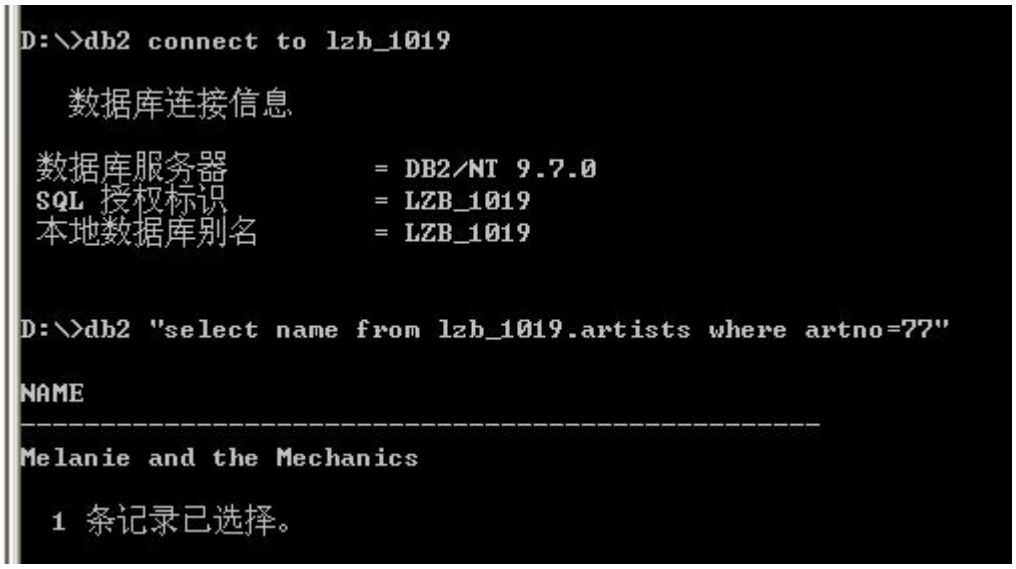
D:\>
```

* RESTORE 成功完成；
* `ROLLFWD_PEND = NO`（没有处于任何暂挂状态，数据库立即可用）。

23. Connect to MUSICDB database.
Issue a SELECT statement against the artists row that was previously updated (ARTNO 77).

Did the value returned to you match what you expected?

Show your work below by printing screen and answer the question.

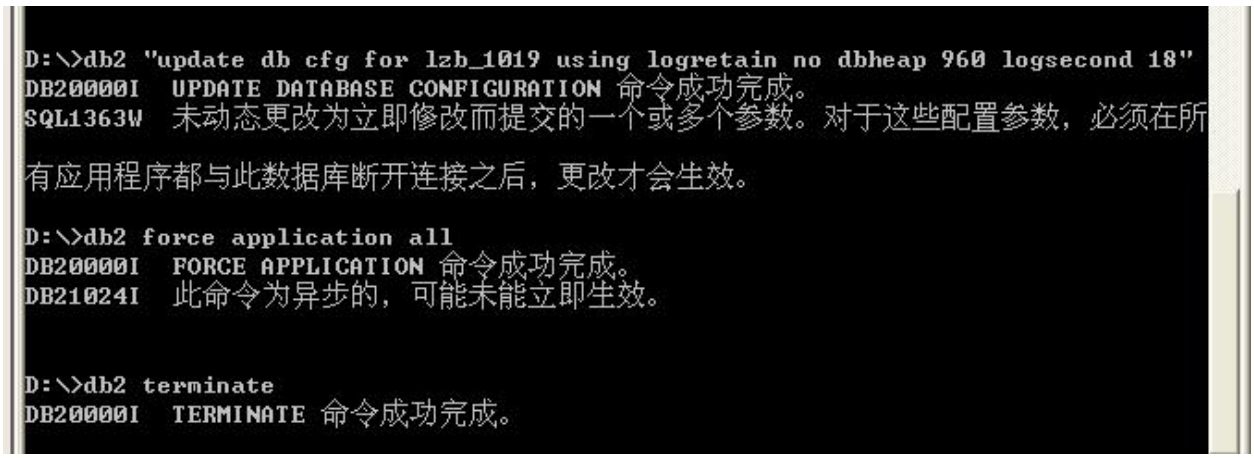


是的。返回的是备份最初是的旧名称。这也证实了，如果在恢复时明确指示不前滚，那么记录在日志中的所有后续修改操作（Melanie 和 Bill）都不会被应用。

Section 4 - Resetting the Environment

1. Execute the cleanup script *clrec*.

Show your work below by printing screen.



2. Examine your db cfg and verify that your database is now configured for circular logging.

Show your work below by printing screen.

